# Data Science

## Gradient Descent, Bayes & Causal Inference



Advanced Modeling

GENERAL ASSEMBLY

# Lesson roadmap



**Welcome + Warm-up**

**Gradient Descent**

**Bayesian Modelling**

**Causal Inference**

**Bringing It Together**

**Bring It Home**

# Learning objectives



1. Fit a model by gradient descent; tune the learning rate
2. Interpret posteriors, credible intervals and priors
3. Estimate causal effects by matching and weighting
4. Report a causal effect with honest uncertainty

# Warm-Up

# Discussion: how would you find out?

A government runs a job-training programme for people struggling to find work.

**Did it raise their earnings?**

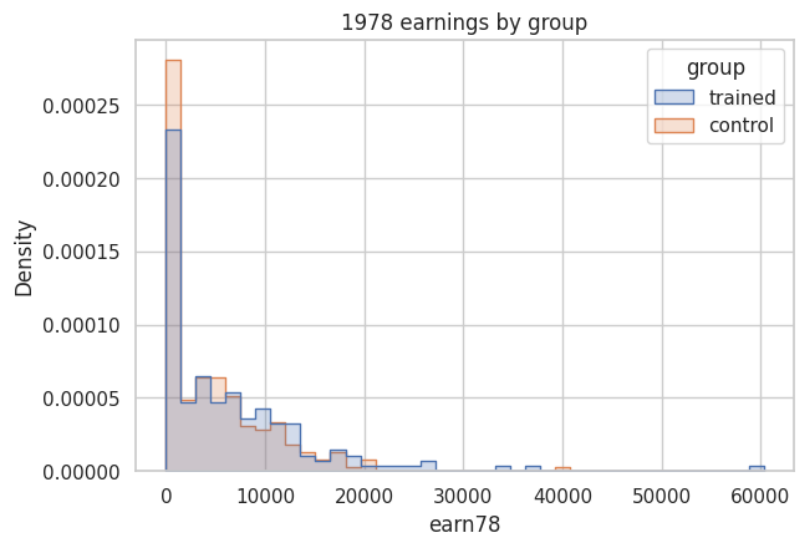
- Could you just compare the earnings of people who took part with people who didn’t?
- What could go wrong with that comparison?
- What would the ideal study look like?

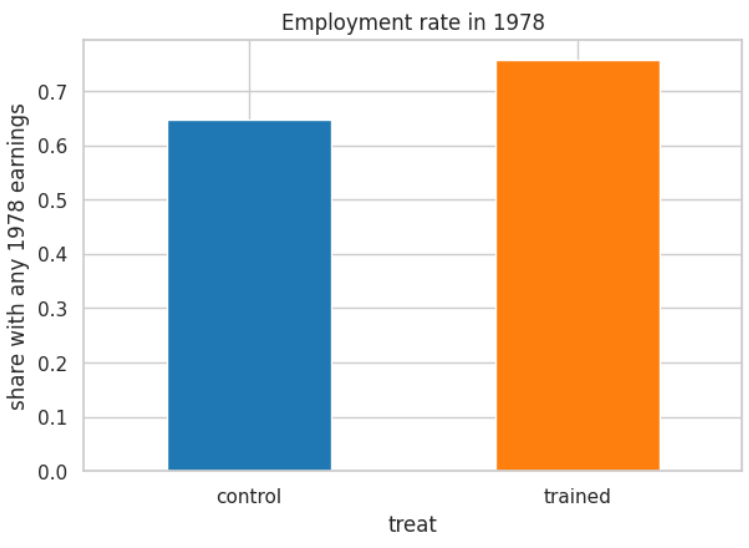
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# Our data: a job-training experiment

National Supported Work (NSW), US, mid-1970s: 9–18 months of supported work. Applicants were **randomly assigned**: 185 trained, 260 controls.

Outcome: **1978 earnings**. Trained people earned **$1,794 more** on average: our **benchmark**.




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# Gradient Descent

# How does a model learn?

Fitting a model = choosing parameters that make a **loss function** as small as possible: **optimisation**.

- **Linear regression:** mean squared error (MSE)
- **Logistic regression:** log-loss (our propensity model, later)
- **Lasso / Ridge:** loss + a penalty (like a Bayesian prior, later)

Sometimes calculus solves it exactly (a **closed form**). Often it can't, so we use **gradient descent**.

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# Derivatives: which way is downhill?

The **derivative** is the slope of the loss at the current point.

- **Positive** slope: the loss rises to the right, so step **left**
- **Negative** slope: step **right**
- **Zero** slope: flat, a possible minimum (or maximum)

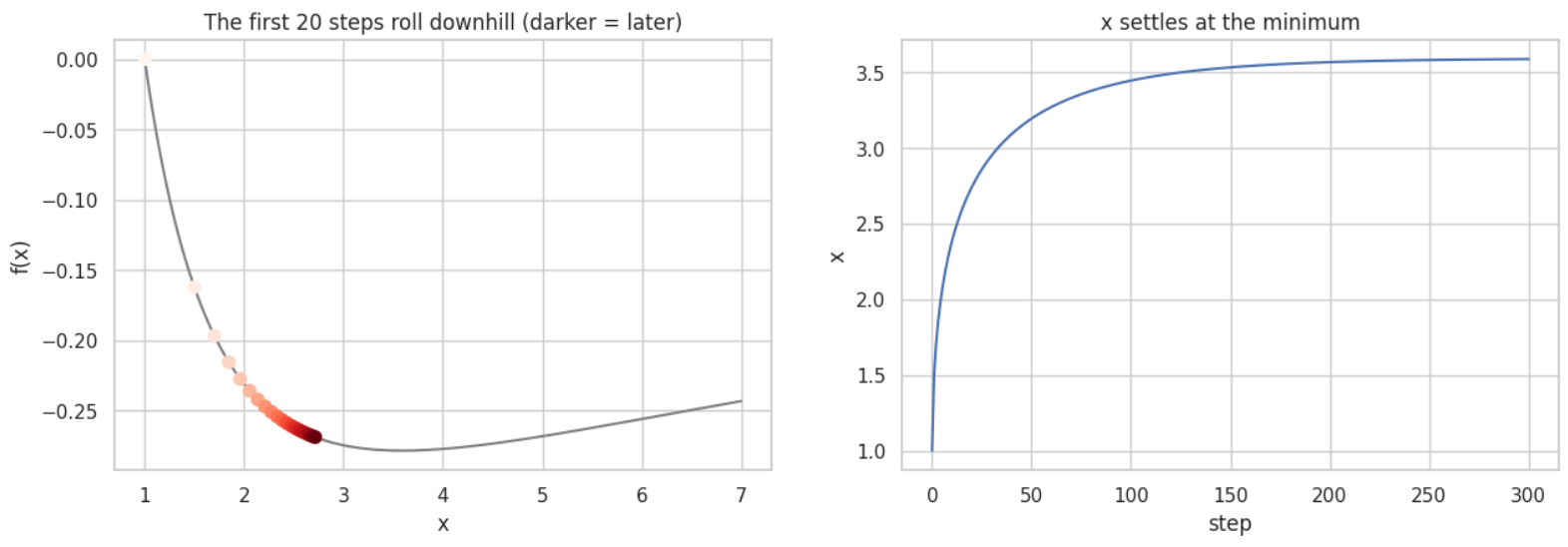
Gradient descent follows the slope downhill, one step at a time.

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# When you can’t solve it by hand

f(x) = –ln(x) / (1 + x): setting the derivative to zero has **no neat solution**.

Gradient descent from x = 1, α = 1: after 300 steps, **x ≈ 3.587**.




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# The gradient descent algorithm

- **Step 0:** start at a random point
- **Step 1:** compute the derivative (the **gradient**)
- **Step 2:** move against it: $\theta \leftarrow \theta - \alpha \times \text{gradient}$
- **Step 3:** repeat steps 1 and 2
- **Step 4:** stop when the steps are small enough

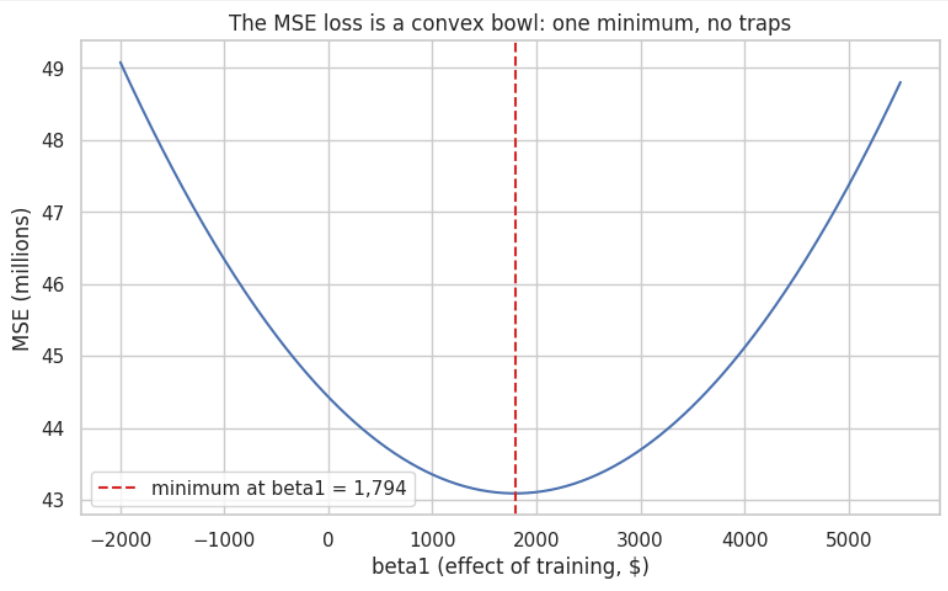
$\alpha$ (alpha) is the **learning rate**: the step size.

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# Our data: the loss is a bowl

Model: earnings = β0 + β1 × trained. The MSE is **convex**: one minimum, at β1 = **$1,794** (the closed form).

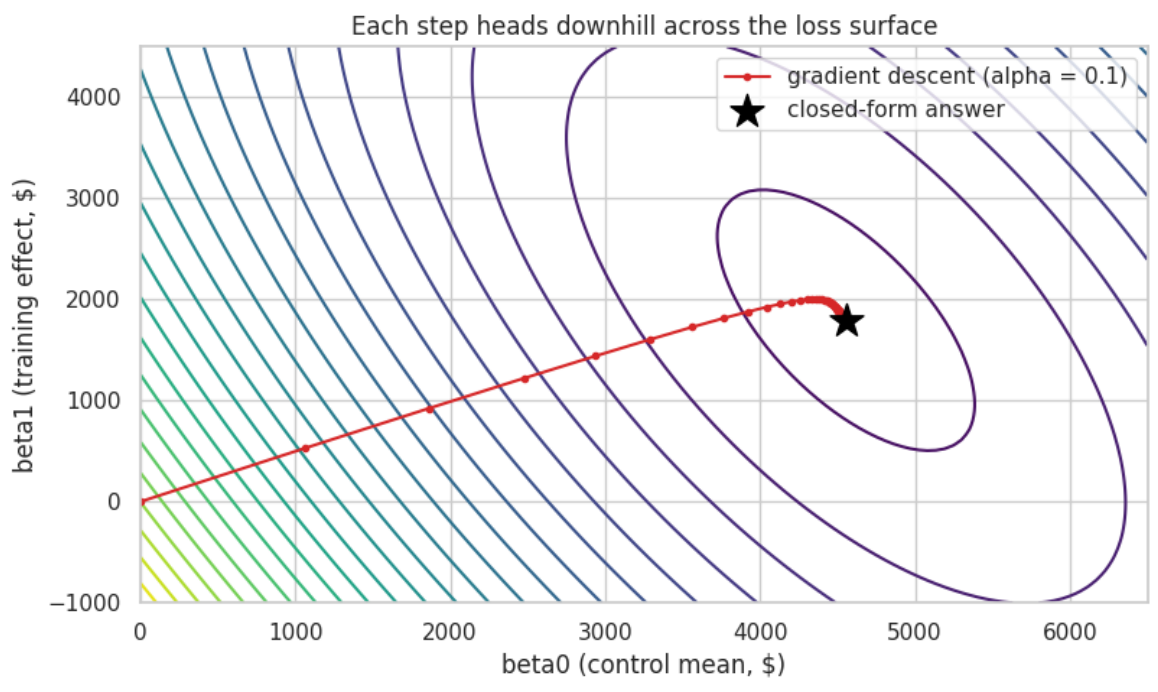
Gradient descent (α = 0.1) lands on **$4,554.8** and **$1,794.3**, exactly the closed-form answer.




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# Watching it learn

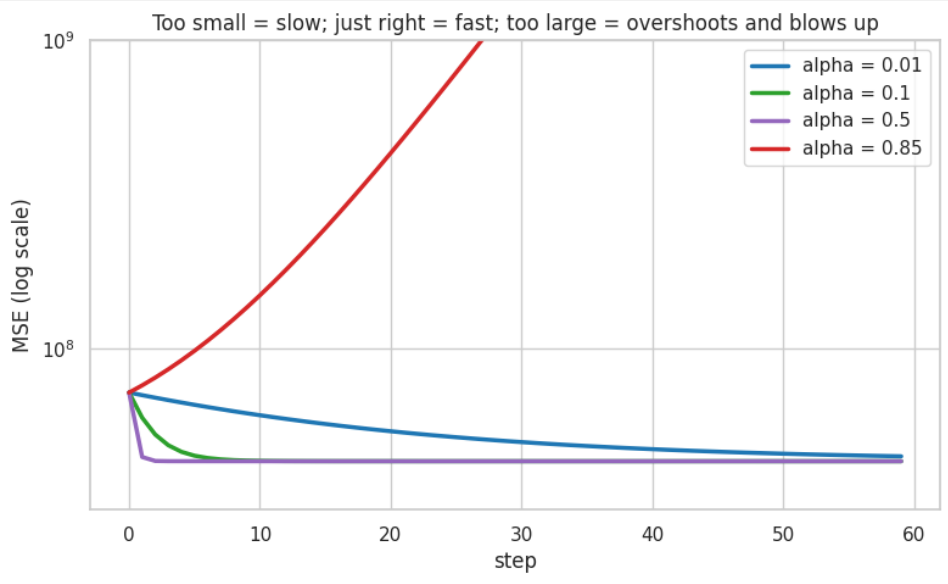
Each step moves **downhill across the loss surface** towards the minimum (★).




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# What can go wrong?

- **Local minima:** avoided when the loss is **convex** (MSE, log-loss)
- **α too large:** overshoots and blows up (α = 0.85)
- **α too small:** painfully slow (α = 0.01)
- **Curve too flat:** tiny gradients, tiny steps

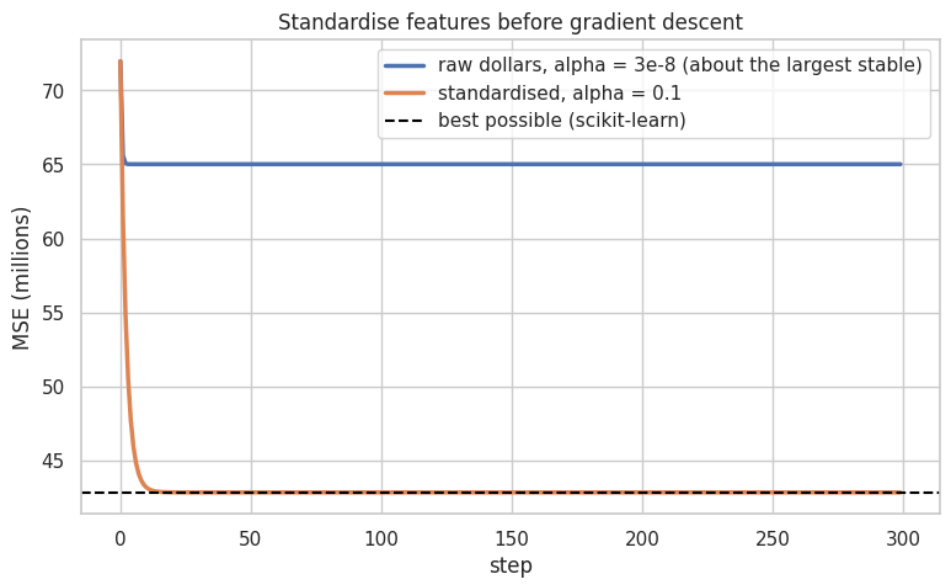



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# Scale your features

Raw dollar earnings make the loss surface long and narrow: the largest **stable** $\alpha$ barely moves the model.

Standardised: converges in a few steps to the best possible fit.




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# Choosing α and smarter optimisers

- **Manually**, or by grid search over a few values
- **Shrink α over time** (learning-rate schedules)
- **Fancier algorithms:** momentum, **Adam** (adaptive moment estimation)
- **Mini-batch / stochastic:** a few rows per step, for big data
- **Newton's method** uses curvature: **7 steps vs 554** on our function. scikit-learn's L-BFGS borrows this idea

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GUIDED WALKTHROUGH:

# Gradient Descent

Let's see it for ourselves! Parts 1–2

→ Open **02a-bayesian-gradient-descent-causal-inference-live.ipynb**




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# Bayesian Modelling

# Two ways to think about uncertainty

**Frequentist:** the true effect is a fixed number; we estimate it and describe how a **method** behaves over repeated samples (confidence intervals, p-values).

**Bayesian:** we describe our uncertainty about the effect with a **probability distribution**, and update it with data.

Both are useful. Bayesian answers read naturally: “there is a 99% probability the programme helped”.

A **frequentist** says: the true effect is a fixed number, and talks about how a method would behave if we repeated the study many times.
A **Bayesian** says: let's describe our uncertainty about the effect with a probability distribution, and update it as data comes in. Both are useful, but Bayesian answers are often easier to explain

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# Priors + data = posterior

**Posterior ∝ likelihood × prior**

- **Prior:** what we believed before seeing the data
- **Likelihood:** how probable the data is for each possible value
- **Posterior:** our updated belief, a whole distribution

Today: two ready-made recipes, **beta–binomial** for rates and **normal–normal** for averages.

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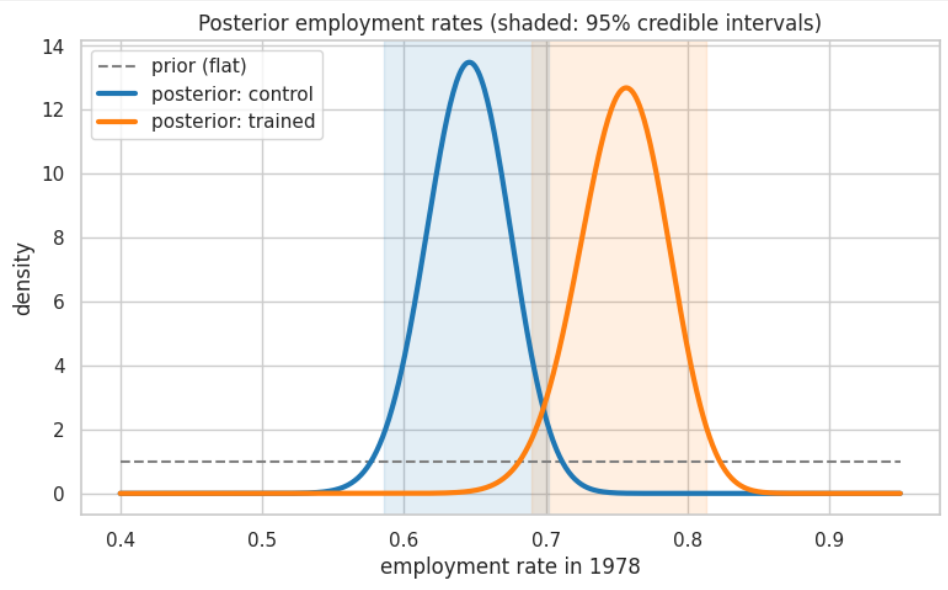
# Example 1: the employment rate

Prior: **Beta(1, 1)**, “any rate is equally plausible”.

Data: 140 of 185 trained and 168 of 260 controls had a job in 1978.

Posterior: **Beta(1 + employed, 1 + not employed)**: just add the counts.

Shaded bands: 95% **credible intervals**.




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# Did training raise employment?

Draw plausible rates from both posteriors and compare them, 100,000 times.

- P(trained rate > control rate) = **99.4%**
- Difference: **+10.9 points** (95% credible interval 2.4 to 19.2)

A direct probability statement that a manager can act on.




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# Credible vs confidence intervals

**95% credible interval:** given the data and prior, there's a **95% probability** the true value is inside.

**95% confidence interval:** 95% of intervals built **by this method** would contain the true value.

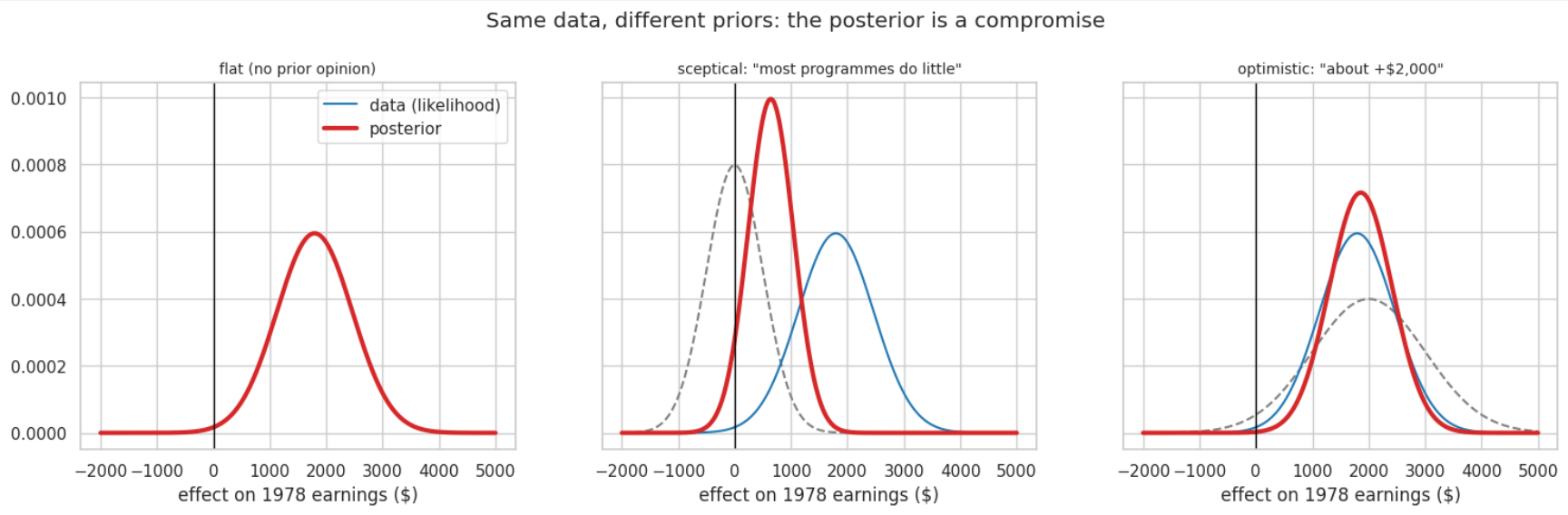
With a flat prior and plenty of data the numbers are often similar; the **meaning** is different.

People usually interpret confidence intervals as if they were credible intervals.

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# Example 2: priors and earnings

Normal prior + the data's estimate ($1,794 ± $671) → normal posterior: a **precision-weighted average**.




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# Prior sensitivity: report it

A confident prior needs a lot of data to overrule. If reasonable priors change the conclusion, **say so**.

| Prior | Posterior mean | 95% credible interval | P(effect > 0) | P(effect > $1,000) |
|---|---|---|---|---|
| Flat (no opinion) | $1,794 | $479 to $3,109 | 99.6% | 88.2% |
| Sceptical (mean 0, sd $500) | $641 | −$145 to $1,426 | 94.5% | 18.5% |
| Optimistic (mean $2,000, sd $1,000) | $1,858 | $766 to $2,950 | 100% | 93.8% |

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# Gradient descent vs Bayes

Gradient descent can minimise the **negative log-posterior** (data loss + a prior penalty). It lands on the posterior's **peak**: **$641** for the sceptical prior.

A normal prior acts like a **Ridge** penalty, pulling estimates towards the prior mean.

But gradient descent gives **one number**; Bayes gives the **whole distribution**, so we can say how sure we are.

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GUIDED WALKTHROUGH:

# Bayesian Modelling

Let's see it for ourselves! Parts 3–4

→ Open **02a-bayesian-gradient-descent-causal-inference-live.ipynb**




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# Break time



# Causal Inference

# Correlation is not causation

A **confounder** affects both who gets the treatment **and** the outcome.

- Background → who joins a training programme
- Background → later earnings

A simple comparison then mixes the **programme's effect** with **pre-existing differences**.

Example: people who join a programme for the unemployed would earn less anyway.

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# Potential outcomes

Each person has two outcomes: **Y(1)** if trained, **Y(0)** if not. Effect = **Y(1) − Y(0)**.

**The fundamental problem:** we only ever see one; the other is **missing**. Rubin: causal inference **is a missing-data problem**.

| Person | Trained? | Y(1): earnings if trained | Y(0): earnings if not | Effect |
|---|---|---|---|---|
| A | yes | $9,930 | ? | ? |
| B | no | ? | $4,200 | ? |
| C | yes | $3,596 | ? | ? |

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# Why randomisation works

A coin flip decides who is trained, so treatment is **unrelated to background**.

The control group's outcomes are then a fair stand-in for the trained group's missing Y(0).

- Check: standardised mean differences (SMD) are small in the experiment

So the simple difference in means, **$1,794**, is a valid causal estimate.

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# LaLonde's test: no experiment

Keep the 185 trained people; replace the controls with **15,992 survey respondents** (not in the programme).

Naive comparison: trained people earned **$8,498 less**.

**The wrong sign!** The experiment says +$1,794.

Why? Look at who the two groups are…

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# The groups were never comparable

Trained people were younger, less educated, more likely Black and unmarried, and earned **far less in 1974–75**.

SMDs of 1.5 to 2.4 (a fair comparison needs |SMD| < 0.1).




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# Fix 1: regression adjustment

Add the confounders to a linear regression alongside the treatment indicator.

Effect “holding background constant”: **$1,066** (benchmark $1,794).

- Much better than –$8,498
- But it assumes straight-line relationships, and leans on survey people who look **nothing like** the trainees

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# The propensity score

**Propensity score** = P(treated | background), estimated with **logistic regression**.

Rosenbaum & Rubin (1983): people with the **same score** have, on average, the same background mix.

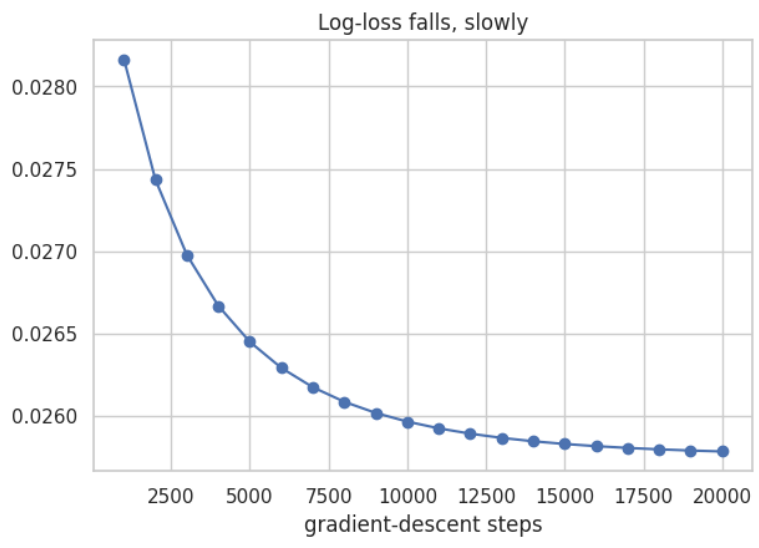
So comparing trained and untrained people with similar scores is like a mini-experiment.

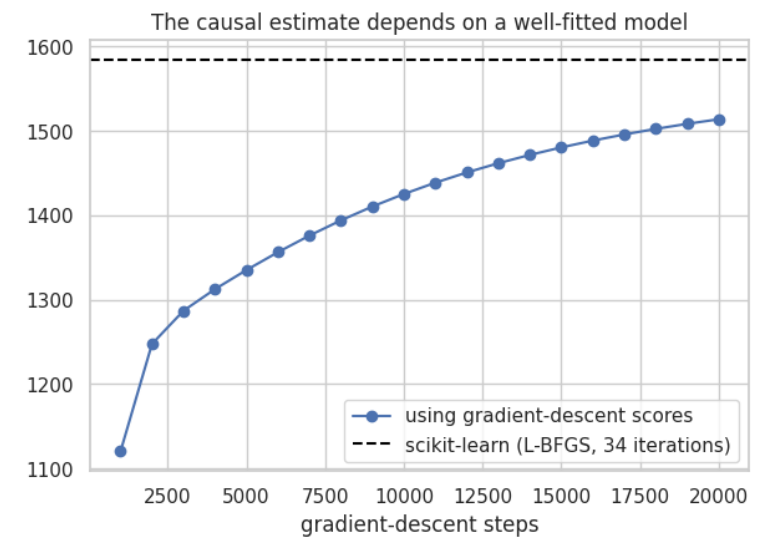
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# Under the hood: an optimiser at work

The propensity model minimises the **log-loss**. Plain gradient descent needs 20,000+ steps and still gives **$1,514**.

scikit-learn's L-BFGS converges in **34 iterations**: **$1,585**. **Check convergence**: it changes the causal answer.



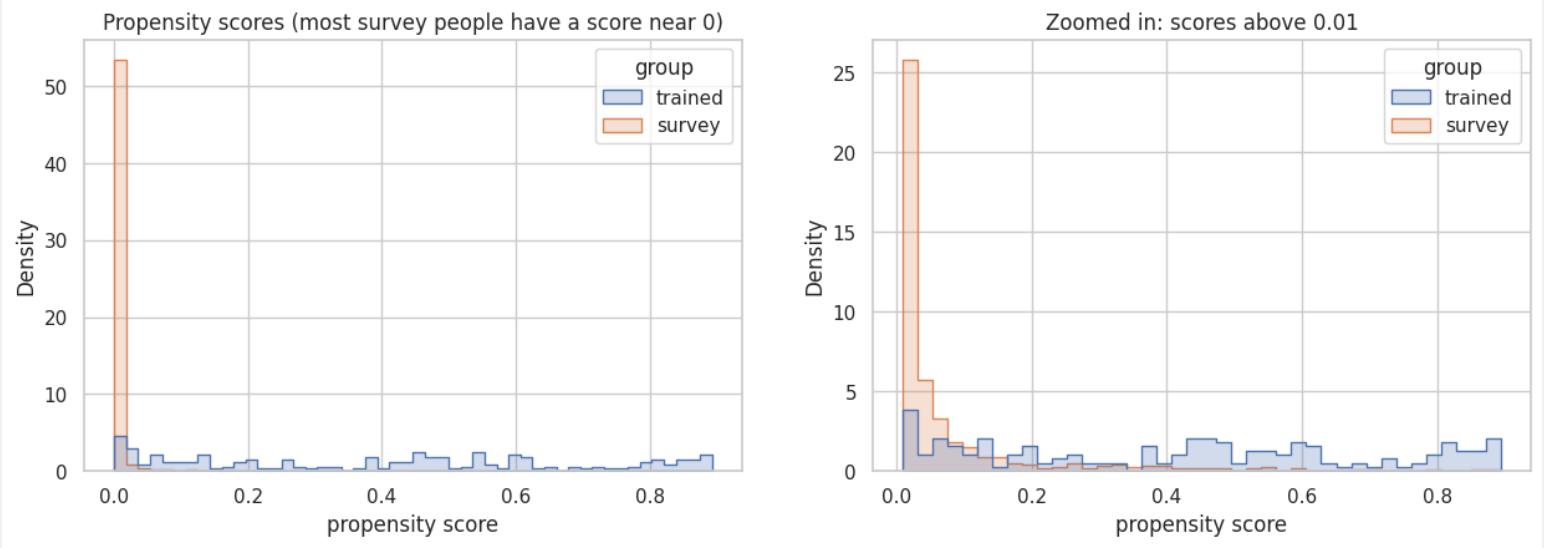




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# Overlap: who is comparable?

Most survey respondents have a score near 0: they look nothing like the trainees.

But several hundred overlap with the trained group: these are the people we need.




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# Fix 2 and 3: matching and weighting

**Matching:** pair each trained person with the survey respondent whose score is closest. Their earnings **fill in the missing Y(0)**.

- Estimate: **$1,642**

**Inverse-probability weighting (IPW):** weight each respondent by score / (1 – score), so the comparison group **looks like the trained group**.

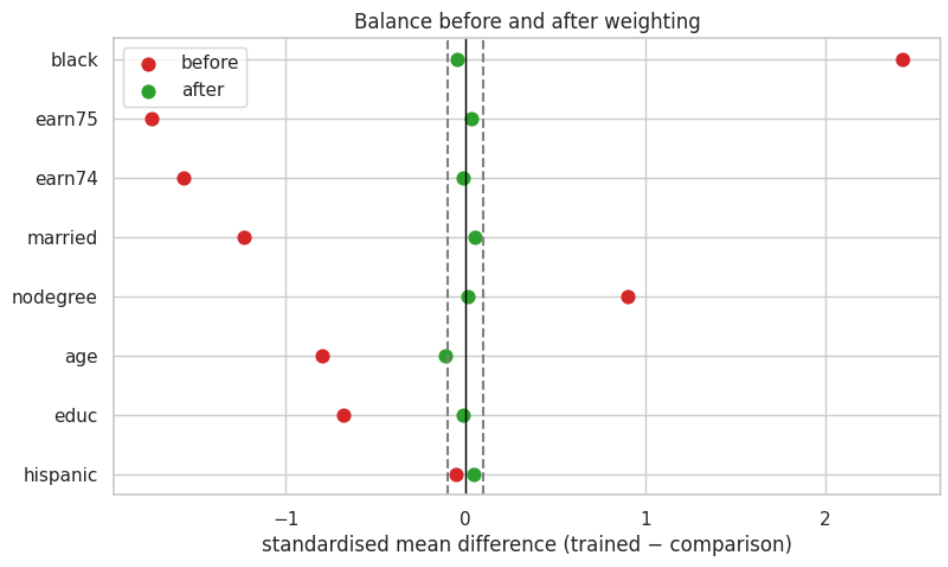
- Estimate: **$1,585** (15,992 people count as about 107 “effective” people)

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# Always check balance

After weighting, every SMD is **within ±0.1**.

That's the evidence that the comparison is now fair.

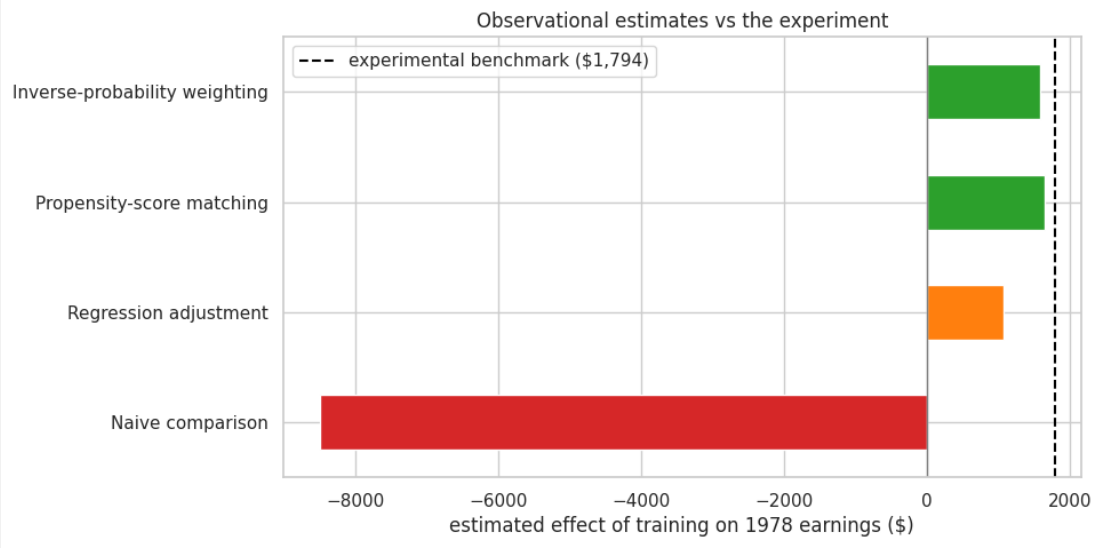



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# The scorecard

Naive −$8,498 → regression $1,066 → matching $1,642 → IPW $1,585.

Experimental benchmark: **$1,794**.




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# The assumptions to state

**1. No unmeasured confounding:** everything that affects both treatment and outcome is measured (here, **previous earnings** are crucial).

**2. Overlap:** comparable people exist in both groups.

**3. No interference:** one person's treatment doesn't change another's outcome.

None can be fully tested: argue for them, and check sensitivity.

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GUIDED WALKTHROUGH:

# Causal Inference

Let's see it for ourselves! Parts 5–6

→ Open **02a-bayesian-gradient-descent-causal-inference-live.ipynb**




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# Bringing It Together

# Causal inference as missing data

Every method fills in the missing potential outcomes:

- **Randomisation:** the control group stands in for everyone
- **Matching:** a similar person stands in
- **Weighting:** a re-weighted group stands in

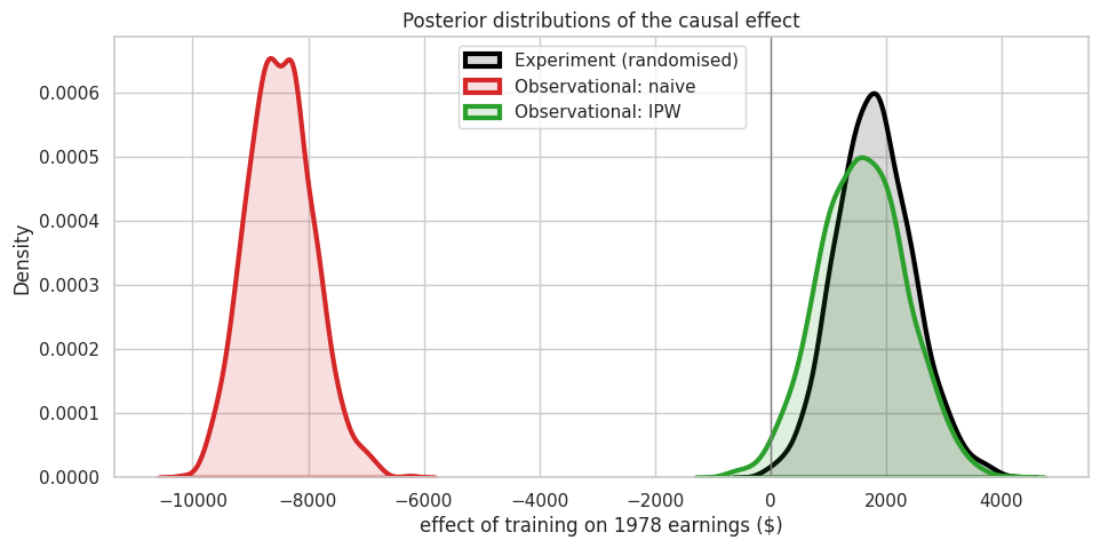
Rubin's Bayesian view: treat the missing outcomes as **unknowns with a posterior**, and carry that uncertainty into the effect.

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# A posterior for the causal effect

Rubin's **Bayesian bootstrap**: random weights for every person, recompute, repeat 1,000s of times.

IPW: mean **$1,589**, 95% CrI $160 to $3,060, P(effect > 0) = **98.7%**. The naive estimate is **confidently wrong**.




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# How to report it

“Weighting the comparison group to match the trainees on age, education, background and previous earnings, we estimate the programme raised 1978 earnings by about **$1,600** per participant (95% credible interval $160 to $3,060; probability of a positive effect about 99%).

This assumes **no unmeasured confounding**; previous earnings are the key adjustment.”

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GUIDED WALKTHROUGH:

# Bringing It Together

Let’s see it for ourselves! Part 7

- Open **02a-bayesian-gradient-descent-causal-inference-live.ipynb**




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# Bring It Home

# Checklist: did X cause Y?

- Define the **treatment**, **outcome** and **population**
- List **confounders** measured **before** treatment
- Naive estimate → **check balance** (SMD, love plot)
- Propensity scores → check **overlap**
- Match or weight → **re-check balance**
- Estimate with **uncertainty**; state **assumptions**; run a **sensitivity** check

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# Key takeaways



**Models learn downhill**

Step against the slope; tune α and scale features



**Know how sure you are**

Prior + data = posterior, e.g. “99% chance it helped”



**Make groups comparable**

Confounders mislead: match or weight, then check balance

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DON'T FORGET:




# Exit tickets


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